HUMAN COMPUTER INTERACTION

Project Report

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1. Introduction

The Crave Corner Cafeteria Ordering System is a web-based platform designed to digitize and streamline the food ordering process at university cafeterias. The system allows students to browse menus from multiple campus canteens, place orders online and collect them without waiting in long queues. This project applies Human Computer Interaction (HCI) principles to ensure the system is usable, efficient and user-friendly for university students.

2. Problem Statement

University students face long waiting queues at campus cafeterias during peak hours such as lunch breaks. This results in:

- Wasted time standing in queues
- Late returns to classes
- Missed meals due to time constraints
- Overcrowding at canteen counters
- No visibility of menu items before reaching the canteen

Target Users

- Primary: University students (age 18-25)
- Secondary: Cafeteria staff and management

Justification

A digital ordering system eliminates physical queues by allowing students to order in advance from their phones or laptops and simply collect when ready. This saves time, reduces crowding and improves the overall campus dining experience.

3. Literature Review

3.1 Background Study

Food ordering systems have become increasingly common in educational institutions worldwide. Universities in developed countries have adopted digital cafeteria systems to improve student experience and reduce operational inefficiencies.

3.2 Existing Systems

- Foodpanda: A food delivery platform allowing users to browse menus and order online. However, it focuses on restaurant delivery, not campus pickup.
- Daraz Food: An e-commerce based food ordering system but not designed for the university cafeteria environment.
- CampusDish (USA): A university dining platform but not available in Pakistani universities.

- Manual Ordering (Current Practice): Students physically visit canteens, wait in queues and order verbally with cash payments only.

3.3 Limitations of Current Solutions

- No existing system specifically designed for Pakistani university cafeterias
- Current manual system causes long queues
- No digital menu visibility before visiting the canteen
- Cash only payments, no digital payment options
- No order tracking or confirmation system
- No student discount management system

4. Methodology

The project followed a User Centered Design (UCD) approach:

Step 1 - Problem Identification: Identified long queues and inefficiency at university cafeterias as the core problem through observation.

Step 2 - User Research: Identified target users as university students who regularly use campus cafeterias.

Step 3 - Conceptual Design: Developed interaction flows, task models and system functionality before interface design.

Step 4 - Interface Design: Created 15 screens using Figma following HCI usability principles.

Step 5 - Usability Evaluation: Conducted usability testing with 3 university students to identify issues and suggest improvements.

Design Approach: Iterative Design — each screen was designed, reviewed and refined based on usability feedback.

5. Design

5.1 Conceptual Design

Interaction Flow: Landing Page → Login/Register → Home → Canteen Menu → Cart → Payment → Order Confirmation

Task Models:

Task 1 - Browsing Menus: User opens website → Views canteen list → Selects canteen → Browses menu items

Task 2 - Placing Order: User adds items to cart → Reviews cart → Proceeds to payment → Confirms order

Task 3 - Tracking Order: User clicks Orders in navbar → Views order status → Tracks preparation progress

System Functionality:

- User registration and login
- Browse 4 campus canteens
- View menu items with prices

- Add items to cart from multiple canteens
- Apply student discount promo code
- Multiple payment methods
- Order confirmation and tracking
- AI Chat-bot assistance
- User profile management

Interaction Styles:

- Menu based navigation (navbar)
- Direct manipulation (add to cart, quantity buttons)
- Form based interaction (login, register, payment)
- Conversational interface (chatbot)

5.2 Interface Design

Tool Used: Figma + Google Stitch

Screens Designed (15 total):

1. Landing Page
2. Login Screen
3. Register Screen
4. Home (Canteens List)
5. Brew & Bites Menu
6. Crust & Co Menu
7. Abbasi & Co Menu
8. Juice Head Menu
9. Cart Screen
10. Payment Screen
11. Order Confirmation
12. My Orders
13. User Profile
14. About Us
15. Chatbot Page

HCI Usability Principles Applied:

1. Visibility of System Status: Order confirmation page shows real time preparation status to keep users informed.

2. Consistency and Standards: Same navbar, footer and color theme (#D4472A orange-red) used across all 15 screens.

3. User Control and Freedom: Back buttons on every page, edit cart option on payment screen.

4. Error Prevention: Form validation on login and register screens, confirm password field prevents wrong passwords.

5. Recognition over Recall: Menu items show images, names and prices clearly so users do not need to memorize anything.

6. Aesthetic and Minimalist Design: Clean peach/white color theme, no unnecessary elements, clear typography throughout.

7. Feedback: Order confirmation screen confirms successful order placement immediately.

8. Flexibility: Promo code field, multiple payment options and special instructions field in cart.

6. Usability Evaluation

Evaluation Type: Formative Evaluation (conducted during design phase to identify and fix issues)

Test Participants: 3 University Students

- User 1: 2nd year Computer Science student
- User 2: 3rd year Business Administration student
- User 3: 1st year Electrical Engineering student

Tasks Given to Users:

1. Register a new account
2. Browse Juice Head menu and add 2 items to cart
3. Apply promo code STUDENT20
4. Complete payment using JazzCash
5. Find your previous orders

Usability Issues Identified

Issue	Problem	Severity	Improvement
Issue 1	Users confused about difference between Login and Register buttons on landing page	Medium	Made Register button more prominent with different color
Issue 2	Users could not find About Us page easily	Low	Moved About Us link to footer which is more conventional
Issue 3	Cart icon not clearly visible on canteen menu pages	Medium	Added item count badge on cart icon to make it more noticeable
Issue 4	Payment page had too many options causing confusion	Medium	Simplified payment method selection with clearer visual hierarchy
Issue 5	Chatbot button not visible on all pages	Low	Added floating orange chat bubble on every page

Results

- All 3 users successfully completed all 5 tasks
- Average task completion time: 4-6 minutes
- Overall satisfaction rating: 4.2 / 5
- Most liked feature: Student discount system
- Most confusing feature: Payment page (initially)

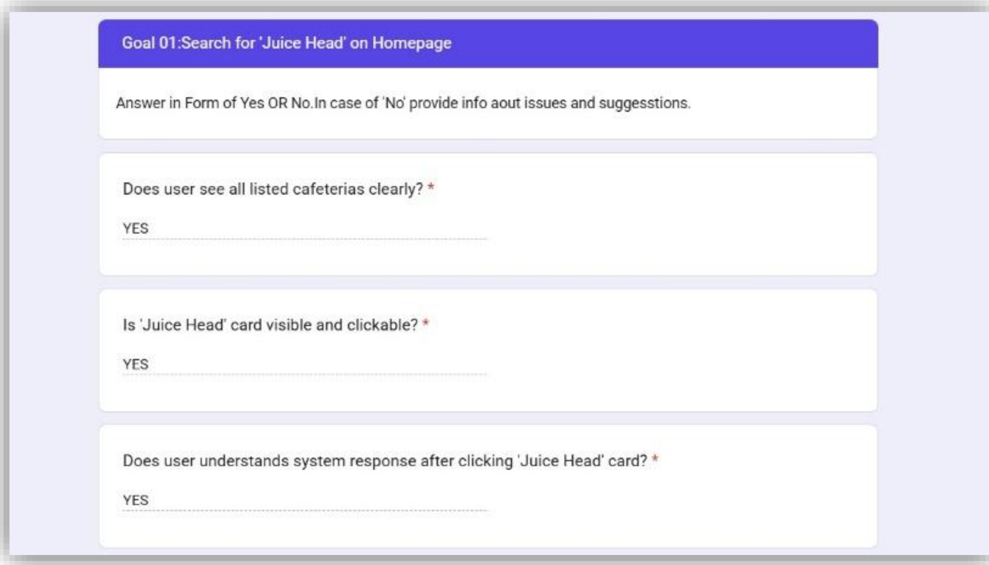
Cognitive Walkthrough

Name: Kulsoom Jahangir.

Role: UI/UX designer

Issue and Suggestion:

Goal 1: No issue


Goal 01:Search for 'Juice Head' on Homepage

Answer in Form of Yes OR No.In case of 'No' provide info aout issues and suggesstions.

Does user see all listed cafeterias clearly? *

YES

Is 'Juice Head' card visible and clickable? *

YES

Does user understands system response after clicking 'Juice Head' card? *

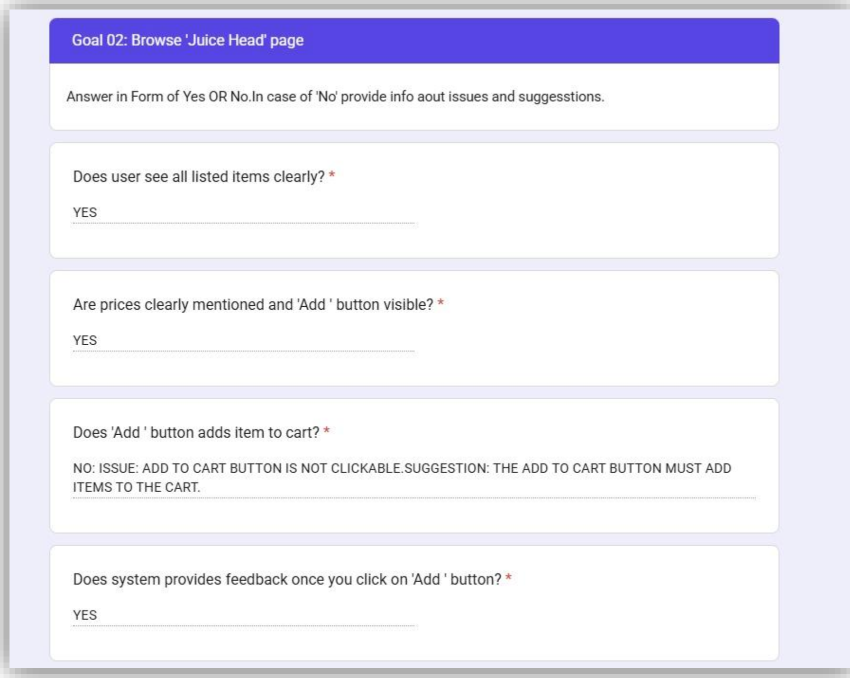
YES

Goal 2: Question 3(issue and suggestion)

NO.

ISSUE: ADD TO CART BUTTON IS NOT CLICKABLE.

SUGGESTION: THE ADD TO CART BUTTON MUST ADD ITEMS TO THE CART.



Goal 02: Browse 'Juice Head' page

Answer in Form of Yes OR No.In case of 'No' provide info aout issues and suggesstions.

Does user see all listed items clearly? *

YES

Are prices clearly mentioned and 'Add ' button visible? *

YES

Does 'Add ' button adds item to cart? *

NO: ISSUE: ADD TO CART BUTTON IS NOT CLICKABLE.SUGGESTION: THE ADD TO CART BUTTON MUST ADD ITEMS TO THE CART.

Does system provides feedback once you click on 'Add ' button? *

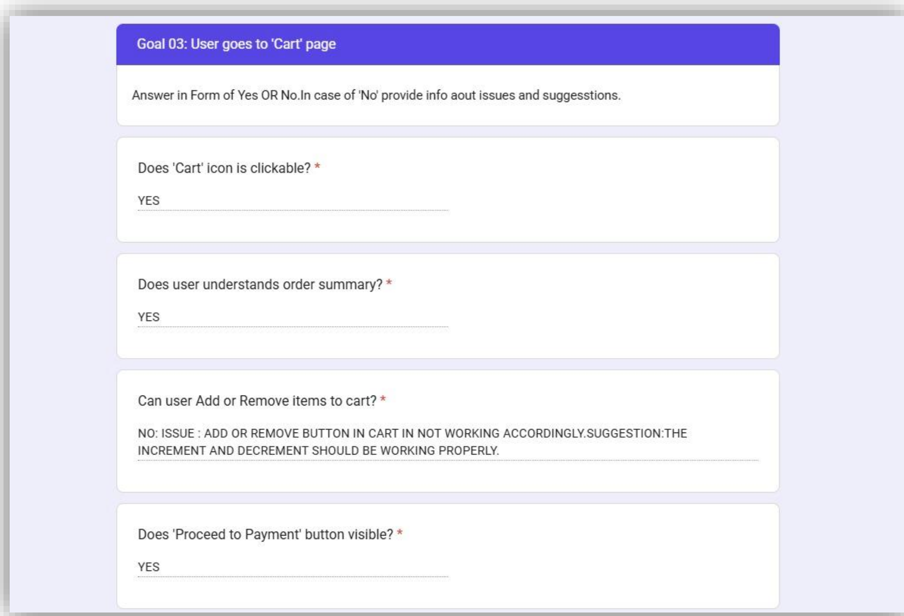
YES

Goal 3: Question 3(issue and suggestion)**.**

NO.

ISSUE: ADD OR REMOVE BUTTON IN CART IN NOT WORKING ACCORDINGLY.

SUGGESTION: THE INCREMENT AND DECREMENT SHOULD BE WORKING PROPERLY.



Goal 03: User goes to 'Cart' page

Answer in Form of Yes OR No.In case of 'No' provide info aout issues and suggestions.

Does 'Cart' icon is clickable? *

YES

Does user understands order summary? *

YES

Can user Add or Remove items to cart? *

NO: ISSUE : ADD OR REMOVE BUTTON IN CART IN NOT WORKING ACCORDINGLY.SUGGESTION:THE INCREMENT AND DECREMENT SHOULD BE WORKING PROPERLY.

Does 'Proceed to Payment' button visible? *

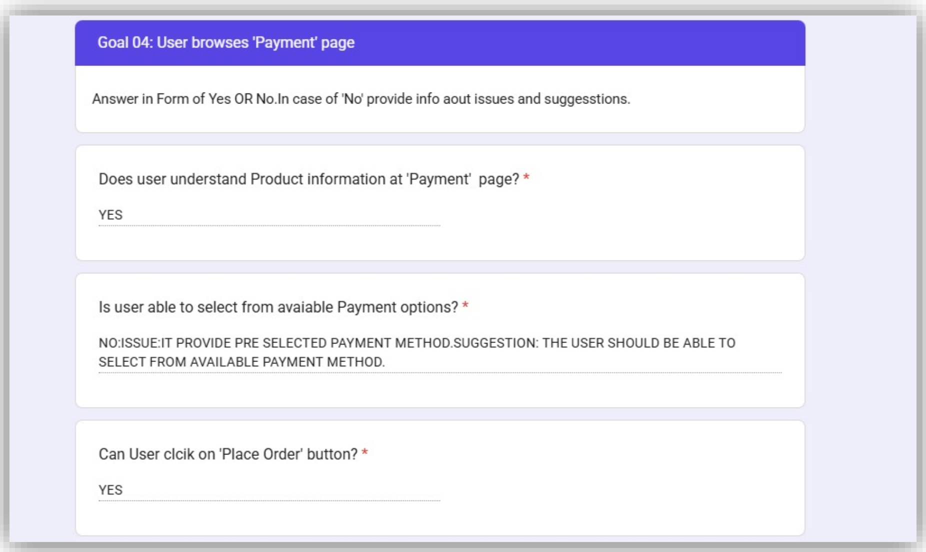
YES

Goal 4: Question 2(issue and suggestion)

NO:

ISSUE: IT PROVIDE PRE SELECTED PAYMENT METHOD.

SUGGESTION: THE USER SHOULD BE ABLE TO SELECT FROM AVAILABLE PAYMENT METHOD.



Goal 5:

Question 2(issue and suggestion)

NO.

ISSUE: THE IS NO REDO OPTION AVAILABLE.

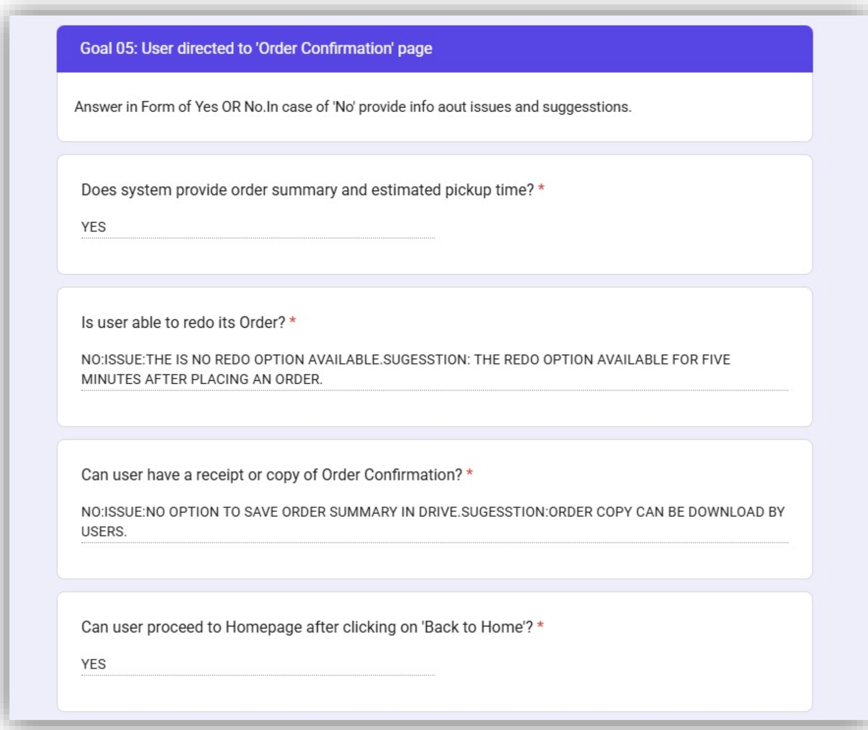
SUGESSTION: THE REDO OPTION AVAILABLE FOR FIVE MINUTES AFTER PLACING AN ORDER.

Question 3 (issue and suggestion)

NO.

ISSUE: NO OPTION TO SAVE ORDER SUMMARY IN DRIVE.

SUGESSTION: ORDER COPY CAN BE DOWNLOAD BY USERS.



Goal 05: User directed to 'Order Confirmation' page

Answer in Form of Yes OR No.In case of 'No' provide info aout issues and suggesstions.

Does system provide order summary and estimated pickup time? *

YES

Is user able to redo its Order? *

NO:ISSUE:THE IS NO REDO OPTION AVAILABLE.SUGESSTION: THE REDO OPTION AVAILABLE FOR FIVE MINUTES AFTER PLACING AN ORDER.

Can user have a receipt or copy of Order Confirmation? *

NO:ISSUE:NO OPTION TO SAVE ORDER SUMMARY IN DRIVE.SUGESSTION:ORDER COPY CAN BE DOWNLOAD BY USERS.

Can user proceed to Homepage after clicking on 'Back to Home'? *

YES

7. Results and Discussion

The Crave Corner system was successfully designed with 15 screens covering the complete user journey from landing page to order confirmation.

Key Findings

- All usability test participants successfully completed assigned tasks
- The consistent color theme and navbar improved navigation efficiency
- Student discount feature (STUDENT20) was well received by all test users
- Chatbot feature added value for first time users
- Multiple payment options (JazzCash, EasyPaisa, Cash) suited Pakistani university students

Improvements Made After Testing

- Simplified payment screen layout
- Added cart badge counter
- Moved About Us to footer
- Made Register button more prominent
- Added floating chatbot bubble on all pages

The system successfully addresses the original problem of long cafeteria queues by providing a complete digital ordering solution that is usable, efficient and tailored to Pakistani university students.

8. Conclusion

This project successfully designed a user-centered cafeteria ordering system called CraveCorner by applying HCI principles throughout the design process. The system solves the real-world problem of long queues at university cafeterias by allowing students to order online from 4 campus canteens and collect when ready.

Key Achievements

- 15 screens designed covering complete user journey
- All major HCI usability principles applied
- Usability testing conducted with 3 users
- Issues identified and improvements implemented
- Student-focused features like discount codes and multiple Pakistani payment methods included

Future Improvements

- Real time order tracking with live updates
- Push notifications for order status
- Loyalty points system for regular customers
- Integration with university student ID system
- Mobile app version for Android and iOS

The project demonstrates that applying HCI principles in the design process results in a more usable, efficient and satisfying user experience for the target audience.

Thank you: